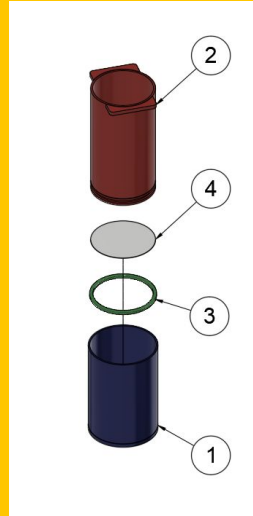


Initial Mockups



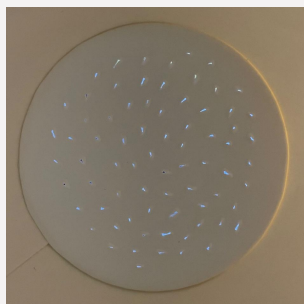
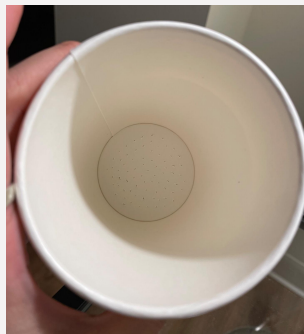
The PressMate

A compact, single serving coffee-brewing device

Initial Mockups

- Main testing goals:
 - Filtration
 - Manufacturability
 - Reduce Costs

Initial Mockups



- Lessons Learned
 - Restrict flow along outer wall
 - Shape of cup cannot be tapered (or barrier must account for a taper)
- Next Steps
 - Create cylindrical, concentric walls

Initial Mockups



- Lessons Learned
 - Fountaining effect upon pressing the plunger
 - Difficult to separate the components
- Next Steps
 -
 - Add tolerance between components
 - Try a mesh instead of punched holes

Initial Mockups



- Lessons Learned
 - No fountaining with mesh
 - Easier user experience with a gap between the plunger and vessel
- Next Steps
 - Machining Mockup

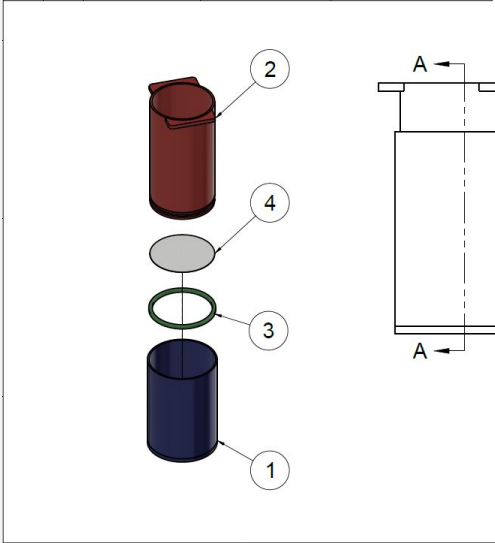
Initial Mockups



- Lessons Learned
 - Cast at Epic is too short
 - Can't use plastic for cup
 - Must be sealed properly
 - Must be easy to pull out plunger, but enough friction to hold cup & plunger together

CAD Drawing

PARTS LIST				
ITEM	QTY	PART NUMBER	DESCRIPTION	MATERIAL
1	1	ASMB_VESS-001	VESSEL ASSEMBLY	STEEL
2	1	ASMB_PLNG-001	PLUNGER ASSEMBLY	STEEL
3	1	SEAL-003_OTS	X PROFILE SEAL	RUBBER
4	1	FLTR-001_OTS	FILTER DISC	STAINLESS STEEL AISI 304



PARTS LIST				
ITEM	QTY	PART NUMBER	DESCRIPTION	MATERIAL
1	1	PLNG_BODY-001_MACH	PLUNGER BODY	STEEL
2	1	PLNG_BOT-001_MACH	PLUNGER BOTTOM	STEEL
3	2	PLNG_TAB-001_MACH	PLUNGER TAB	STEEL

